4.4.1 Wave motion

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	progressive waves; longitudinal and transverse waves	
(b)	**(i)** displacement, amplitude, wavelength, period, phase difference, frequency and speed of a wave	
	(ii) techniques and procedures used to use an oscilloscope to determine frequency	**PAG5**
(c)	the equation $f = \frac{1}{T}$	
(d)	the wave equation $v = f\lambda$	
(e)	graphical representations of transverse and longitudinal waves	
(f)	**(i)** reflection, refraction, polarisation and diffraction of all waves	Learners will be expected to know that diffraction effects become significant when the wavelength is comparable to the gap width.

(ii) techniques and procedures used to demonstrate wave effects using a ripple tank

(f) (iii) techniques and procedures used to observe polarising effects using microwaves and light **PAG5**

(g) intensity of a progressive wave; $I = \frac{P}{A}$; intensity $\propto$ (amplitude)^2.